WebClient Computing

HTML Basics

Department of Computer Science

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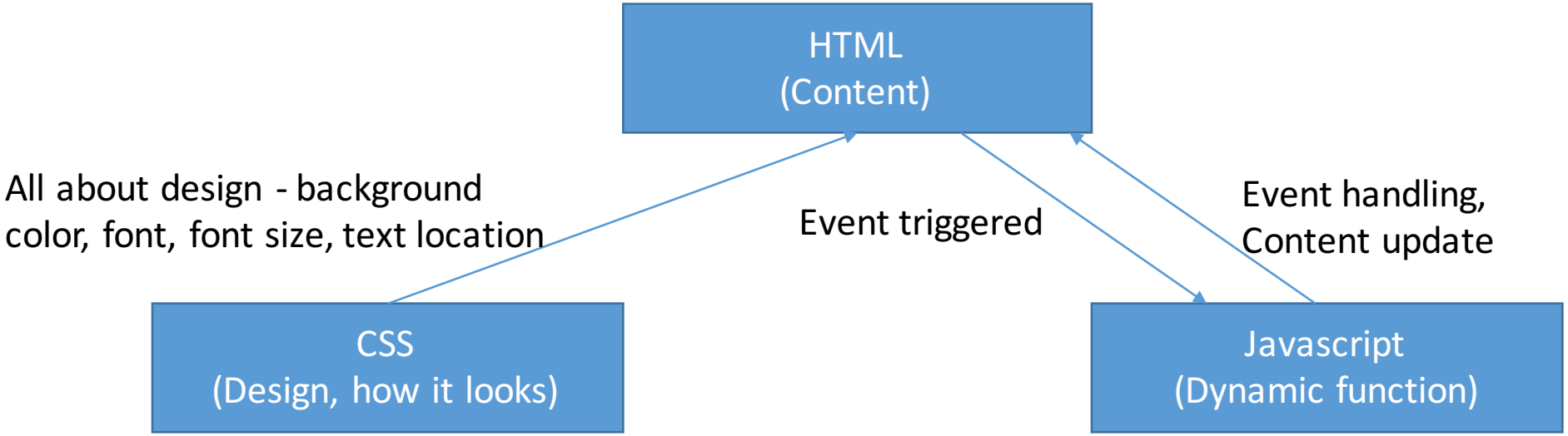
Announcement

- HTML books recommendation will be made by the next Thursday
- The HTML file should be stored in plain text
  - Windows NotePad (메모장) should be OK
  - Vim editor is OK
  - Mac TextEdit embeds rich text features and did not render correctly
- Slight change to the Office-hour
  - Tues. 13:30 ~ 14:30
  - Wed. 10:00 ~ 11:00
  - Fri. 9:30 ~ 10:30
- Calendar address is on the slack
  - http://bit.ly/1pr4fFP

HTML, CSS, and Javascript



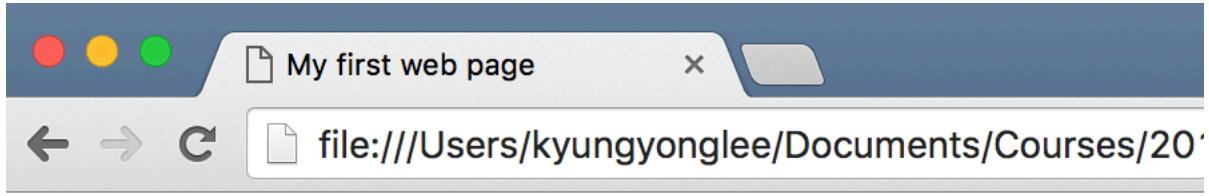
- HTML – Hyper Text Markup Language
  - House itself
- CSS - Painting the house and decoration
- Javascript – rich features, electricity on/off using a smartphone

Quotation and Abbreviation

- <q>
  - Quotes a short sentence – This is <q>short quotation</q>
- <blockquote>
  - Relatively long quotation – Indents the messages
- <abbr>
  - Defines abbreviation or an acronym
  - 70th year of <abbr title=”Kookmin University">KMU</abbr>
  - Useful for search engine to index a page

Examples

```
<!DOCTYPE html>
<html>
<head>
  <title>My first web page</title>
</head>
<body>
  This is <q>short quotation</q>
  <p>Here is a quote from WWF's website:</p>
  <blockquote cite="http://www.worldwildlife.org/who/index.html">
  For 50 years, WWF has been protecting the future of nature.
  </blockquote>
  70<sup>th</sup> year of <abbr title="Kookmin University">KMU</abbr>
</body>
</html>
```



This is "short quotation"

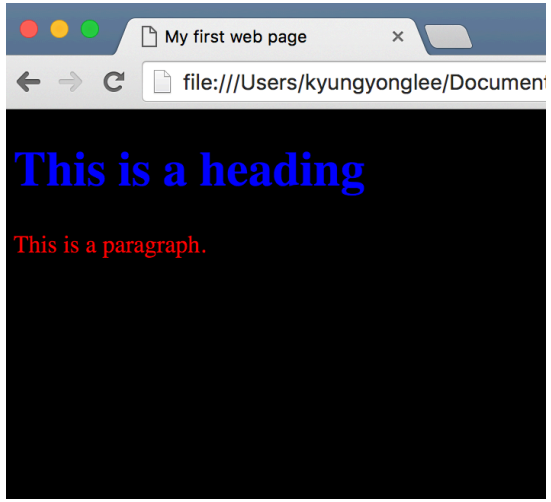
Here is a quote from WWF's website:

> For 50 years, WWF has been protecting the future of nature.

70th year of KMU

HTML Background and Text Color

- Style attributes – setting the style of an HTML element
  - Syntax – style="property:value;"
    - Property and value is in CSS context - not covered in this course
- background-color
  - Sets the background color of a page
- color
  - Sets text color in the HTML element




```
<!DOCTYPE html>
<html>
<head>
  <title>My first web page</title>
</head>
<body style="background-color:black;">
<h1 style="color:blue;">This is a heading</h1>
<p style="color:red;">This is a paragraph.</p>
</body>
</html>
```

Color Definition in HTML

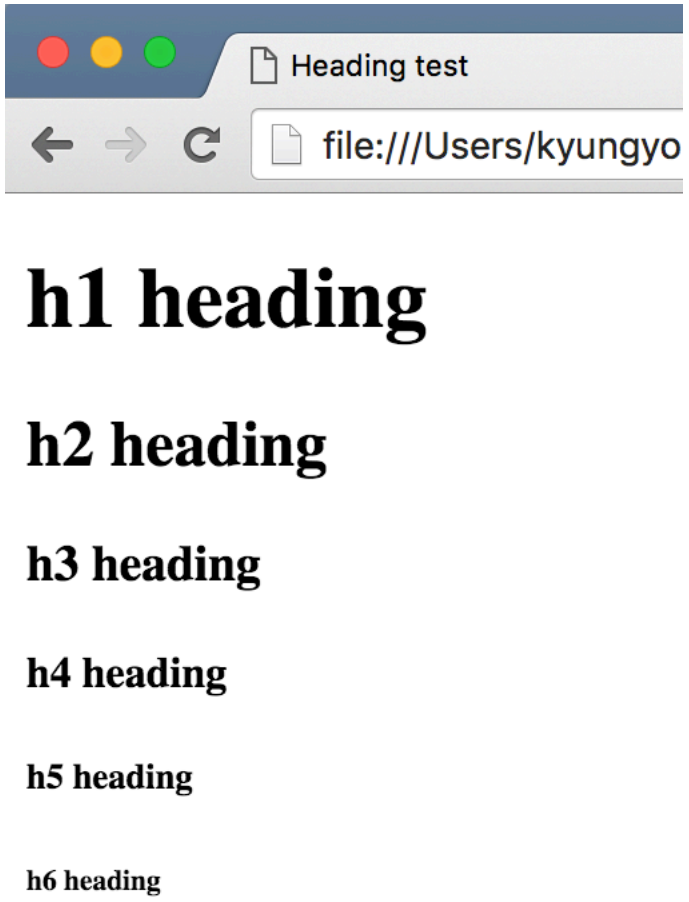
- Modern browsers support 140 colors
  - http://www.w3schools.com/colors/colors_names.asp

Color Name	HEX	Color	Shades	Mix
AliceBlue	#F0F8FF		Shades	Mix
AntiqueWhite	#FAEBD7		Shades	Mix
Aqua	#00FFFF		Shades	Mix
Aquamarine	#7FFFD4		Shades	Mix
Azure	#F0FFFF		Shades	Mix
Beige	#F5F5DC		Shades	Mix
Bisque	#FFE4C4		Shades	Mix
Black	#000000		Shades	Mix
BlanchedAlmond	#FFEBCD		Shades	Mix
Blue	#0000FF		Shades	Mix

HTML Headings

- Organizing contents hierarchically
  - Recommended to use in order
  - h1, h2, h3, h4, h5, h6
  - Space is automatically added after heading
  - Search engine use headings to **index** the structure of the content

```html
<!DOCTYPE html>
<html>
<head>
  <title>Heading test</title>
</head>
<body>
  <h1>h1 heading</h1>
  <h2>h2 heading</h2>
  <h3>h3 heading</h3>
  <h4>h4 heading</h4>
  <h5>h5 heading</h5>
  <h6>h6 heading</h6>
</body>
</html>
```



HTML Fonts, Text Size, and Alignment

- font-family
  - Defines the font to be used for an HTML element
- font-size
  - Defines the text size
- text-align
  - Sets horizontal text alignment for an HTML element

Font Example

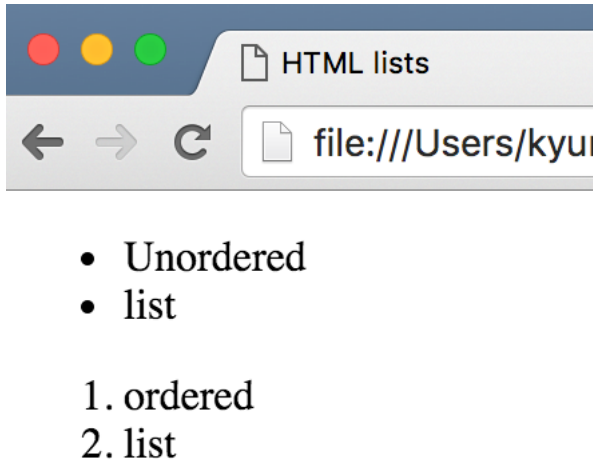
```
<!DOCTYPE html>
<html>
<head>
  <title>My first web page</title>
</head>
<body>
<h1 style="font-family:verdana;font-size:300%;text-align:center;">This is a heading</h1>
<p style="font-family:courier;font-size:160%;text-align:right;">This is a paragraph.</p>

</body>
</html>
```

HTML Lists

- Represents a set of lists in the content
  - ul – unordered list (usually expressed with bullets)
  - ol – ordered list (usually expressed with incremental numbers)
  - li - an element in the list

```html
<!DOCTYPE html>
<html>
<head>
  <title>HTML lists</title>
</head>
<body>
  <ul>
    <li>Unordered</li>
    <li>list</li>
  </ul>
  <ol>
    <li>ordered</li>
    <li>list</li>
  </ol>
</body>
</html>
```



Lists Bullet Shape

- Unordered list style
  - list-style-type:disc - entry mark with bullet (default)
  - list-style-type:circle - entry mark with circle
  - list-style-type:square - entry mark with square
- Ordered list attribute
  - type="1" : numbered 1, 2, 3, …
  - type = "A" or "a": numbered A, B, C, ...
  - type = "I" or "i" : number in roman number in upper or lower case

List Example

```
<!DOCTYPE html>
<html>
<head>
  <title>HTML lists</title>
</head>
<body>
  <ul style="list-style-type:circle">
    <li>Coffee</li>
    <li>Tea</li>
    <li>Milk</li>
  </ul>
  <ol type="A">
    <li>Coffee</li>
    <li>Tea</li>
    <li>Milk</li>
  </ol>
</body>
</html>
```

- Coffee
- Tea
- Milk

A. Coffee
B. Tea
C. Milk

Nested Lists

- Lists can be nested

```
<!DOCTYPE html>
<html>
<body>

<h2>A Nested List</h2>

<ul>
  <li>Coffee</li>
  <li>Tea
    <ul>
    <li>Black tea</li>
    <li>Green tea</li>
    </ul>
  </li>
  <li>Milk</li>
</ul>

</body>
</html>
```

A Nested List

- Coffee
- Tea
  - Black tea
  - Green tea
- Milk

HTML Time Elements

- Express time information in the page
  - <time>2016-03</time>
  - <time>2016-03-07</time>
  - <time>13:49</time>
  - <time>13:49:12</time>
  - <time datetime="2016-02-14">Valentines day</time>
- Time element does not render special. It simply lets browser to know it is information about time

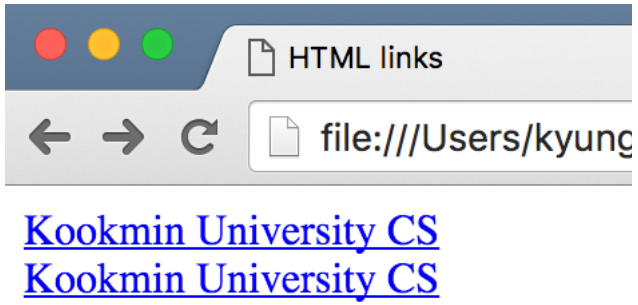
HTML – Make a Link to Other Page

- Hyper Text
  - a structured text that uses links to navigate through different pages
- WWW – World Wide Web
  - Interlinked by hyper text
- Anchor tag <a> represents a link
- Destination is added as an attribute within <a> tag
  - <a href="cs.kookmin.ac.kr">Kookmin Univ. Computer Science</a>

HTML Link Example

- To open the link in a new window add target="_blank" attribute

```
<!DOCTYPE html>
<html>
<head>
  <title>HTML links</title>
</head>
<body>
  <a href="http://cs.kookmin.ac.kr">Kookmin University CS</a><br>
  <a href="http://cs.kookmin.ac.kr" target="_blank">Kookmin University CS</a>
</body>
</html>
```



HTML – Link within a Page

- Within a page, we can assign ID to an entry and link to it

```
<!DOCTYPE html>
<html>
<head>
  <title>HTML page links</title>
</head>
<body>
  <p id="bookmark-1">This is a new paragraph</p>
  <br><br><br><br><br><br><br><br><br><br><br><br>
  <br><br><br><br><br><br><br><br><br><br><br><br>
  <br><br><br><br><br><br><br><br><br><br><br><br>
  <br><br><br><br><br><br><br><br><br><br><br><br>
  <br><br><br><br><br><br><br><br><br><br><br><br>
  <a href="#bookmark-1">jump in a page</a>
</body>
</html>
~
```

References

- http://www.w3schools.com/html/ - HTML tutorial
- http://www.htmldog.com/guides/html/
- http://fromyou.tistory.com/581